

# Papineni Sai Sharan

AI/ML | Cybersecurity

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## Summary

Computer Science Engineer with professional experience in database performance and production reliability, along with hands-on experience in **Python, machine learning, and cybersecurity research**. Experienced in **ML/DL model development and evaluation, ensemble learning, and Explainable AI using SHAP and LIME**. Research experience in **cross-domain threat detection across IIoT and Darknet environments**.

## Experience

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| <b>Associate Database Engineer, Mydbops</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 04/2025 – Present<br>Bengaluru, India |
| <ul style="list-style-type: none"><li>• Monitor database health across <b>MySQL, PostgreSQL, TiDB, and MongoDB</b>.</li><li>• Analyze <b>query performance, execution plans, replication, resource usage, and storage</b> to identify performance issues.</li><li>• Identified and resolved database query performance bottlenecks through detailed execution plan analysis, resulting in a <b>20% improvement in system responsiveness</b>.</li><li>• Improved critical SQL query performance through execution plan analysis, reducing <b>daily processing time by 4 hours</b>.</li><li>• Recommended indexing, query, and table maintenance improvements based on performance analysis and troubleshoot production database issues.</li></ul> |                                       |

## Education

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| <b>Bachelor of Engineering in Computer Science and Engineering,</b><br><i>Vivekananda Institute of Technology</i> | 12/2020 – 10/2024<br>Bengaluru |
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## Skills

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| <b>Programming</b><br>Python                                                                                       | <b>AI / Machine Learning</b><br>Supervised Learning, Deep Learning, Ensemble Models, Model Evaluation, Benchmarking |
| <b>Explainable AI</b><br>SHAP, LIME                                                                                | <b>Databases</b><br>MySQL, PostgreSQL, TiDB, MongoDB                                                                |
| <b>Performance &amp; Monitoring</b><br>Query Analysis, Execution Plans, Replication Monitoring, Query Optimization | <b>Systems &amp; Tools</b><br>Linux, SSH, System Monitoring, Git, GitHub                                            |
| <b>Cloud</b><br>AWS (EC2, RDS, Aurora)                                                                             | <b>Cybersecurity</b><br>Intrusion Detection, Threat Detection, Network Security                                     |

## Awards

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| <b>World Record Participant — Tyagaraja Aradhana Pancharatna Kritis, ICMDA</b><br>Participated in the world record-setting rendition of the Pancharatna Kritis of Saint Tyagaraja on January 7, 2026. | 07/01/2026 |
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## Publications

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### **Cross-Domain Threat Detection: XAI and Ensemble ML for IIoT and Darknet**

22/01/2026

**Scenarios**, Sathwik Narkedimilli, Papineni Sai Sharan, Vidya A., Rajendra Hegadi  
*IEEE 4th International Conference on Data, Decision and Systems (ICDDS 2025)*

- Developed a **cross-domain intrusion detection framework** using the **CIC-APT-IIoT-2024** and **CIC-Darknet2020** datasets.
- Implemented and evaluated **Machine Learning, Deep Learning, and Ensemble models** for threat detection.
- Applied **SHAP and LIME** to interpret model predictions and identify important features.
- Performed **data preprocessing, feature engineering, and model evaluation** using accuracy, precision, recall, and F1-score.
- Achieved **99% accuracy on CIC-APT-IIoT-2024** and **98% on CIC-Darknet2020** with high-performing models and ensemble approaches.

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